Predicting Sediment Particle Size Distributions with Machine Learning

UE24CS352A – Machine Learning | Mini-Project Write-up
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GitHub: https://github.com/ArundhathiK29/ml-sediment-mini-project

1. Problem Statement

Suspended marine sediment (mud) controls carbon sequestration, contaminant transport and coastal erosion, and sediment-transport models need the **particle settling velocity** as a key input. Settling velocity depends on particle size, but size changes within seconds as particles aggregate and break up (*flocculation*), and existing semi-empirical flocculation models are tuned to specific sites and do not generalise. Following Egan, *"Machine learning for predicting sediment particle size distributions"*, we ask: **can a data-driven regression model predict the median particle diameter d50 (µm) and the particle-size-distribution variance σ² (µm²) from measured water-quality, biological and hydrodynamic variables?** Success is measured by R^2 between predicted and true values (supported by MAE and RMSE).

2. Dataset

The paper's field data (1648 hourly samples from South San Francisco Bay, measured with LISST, ECO-FL and ac-9 instruments) is not available, so we use a **synthetic dataset of 5000 samples** (seed 42, no missing values) with the same 8 inputs and 2 targets. Inputs are drawn independently and uniformly. The targets are *illustrative* functions of the inputs (linear terms, S×T and u_b×S interactions, and a u_b^2 term) plus Gaussian noise (σ = 12 µm for d50, 8 µm² for variance); they are **not** validated sediment physics. Targets: d50 mean 174.4 µm (std 46.5, range 48.7–370.9); σ² mean 143.0 µm² (std 38.2, range 44.6–272.9). Because the noise is known we can compute the **noise ceiling**, the best R^2 any model can reach: **0.930 for d50 and 0.958 for σ²**.

Feature (symbol)	Range	Corr. with d50	Corr. with σ²	Role in model
Salinity (S, ppt)	5 – 35	+0.51	+0.74	Selected
Temperature (T, °C)	5 – 30	+0.56	+0.47	Selected
Wave-induced velocity (u_b, m/s)	0.01 – 0.50	−0.43	+0.39	Selected
Particle refractive index (n_p)	1.05 – 1.25	+0.23	+0.14	Selected
Mean tidal velocity (ū, m/s)	0.01 – 1.00	+0.02	+0.09	Dropped
Chlorophyll-a (chl-a)	0.1 – 20	+0.07	+0.03	Dropped
Algal peak ratio (a676/a650)	0.70 – 1.50	0.00	+0.01	Dropped
Detrital peak ratio (a450/a676)	0.50 – 2.00	+0.01	−0.02	Dropped

Table 1: input features, ranges, Pearson correlation with each target (full dataset) and whether the feature was kept after selection.

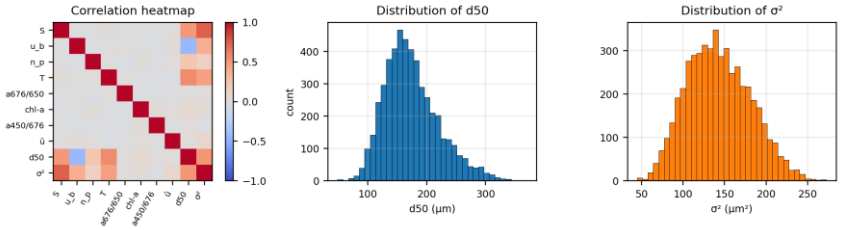


Fig. 1: correlation heatmap and target distributions. Targets are unimodal and mildly right-skewed; d50 and σ² correlate at +0.52.

EDA observations. Salinity and temperature correlate positively with both targets. Wave velocity correlates *negatively* with d50 (−0.43) but positively with σ² (+0.39), because the u_b×S interaction outweighs its direct term for d50. The four biological/optical and tidal features are essentially uncorrelated with the targets, and features are mutually independent by construction.

3. Approach

1. **Split.** Hold out a 20 % test set (1000 samples), used once at the end. All feature selection and tuning use only the remaining 4000 training samples.
2. **Feature selection.** Fit a separate Random Forest (RF) per target on 10 random 70/30 splits of the training set and average two importance measures: impurity (mean decrease in impurity) and permutation importance. Keep the **top 4** features by mean importance, as in the paper.
3. **RF tuning.** Choose the number of trees from the out-of-bag (OOB) R^2, an internal validation estimate that needs no extra validation set.
4. **SVR tuning.** RBF-kernel Support Vector Regression with a grid search over C ∈ {1, 10, 100}, ε ∈ {0.1, 0.3}, γ ∈ {scale, 0.1} using 3-fold CV. Inputs and target are standardised *inside* the pipeline (no leakage, and ε is unit-free).
5. **Evaluation.** Report R^2, MAE and RMSE on the test set, confirm with 5-fold CV on all 5000 samples, and compare against the noise ceiling and the all-8-feature RF.

4. Implementation Overview

Python 3 with NumPy, pandas, scikit-learn and matplotlib, in one notebook (**ml_mini_project.ipynb**) that runs top to bottom with no external uploads: data generation → EDA → split → importance → tuning → evaluation → saved results (**results/**: CSV tables, test predictions, trained models; **data/**: dataset). Models: **RandomForestRegressor** (one per target) and **SVR** wrapped in **TransformedTargetRegressor** + **StandardScaler**; tuning with **GridSearchCV** and the OOB score. All random states are fixed to 42 (reproducible).

Model	Target	Tuned setting	How chosen	Tuning score
Random Forest	d50	100 trees	OOB R^2 flat from ≈20 trees; 100 = smallest within 0.002 of best	OOB R^2 = 0.903
Random Forest	σ²	100 trees	same rule	OOB R^2 = 0.933
SVR (RBF)	d50	C = 10, ε = 0.1, γ = 0.1	3-fold grid search (18 combinations)	CV R^2 = 0.9196
SVR (RBF)	σ²	C = 1, ε = 0.1, γ = 0.1	3-fold grid search (18 combinations)	CV R^2 = 0.9453

Table 2: final hyper-parameters. The σ² SVR optimum C = 1 lies at the lower edge of the grid; its score is almost unchanged for larger C.

5. Results

Target	Model	Features	Test R^2	MAE	RMSE	5-fold CV R^2 (mean ± std)	Noise ceiling
d50 (µm)	Random Forest	all 8	0.914	10.89	13.55	–	0.930
d50 (µm)	Random Forest (tuned)	top 4	0.909	11.24	13.95	0.905 ± 0.005	0.930
d50 (µm)	**SVR (tuned)**	top 4	**0.921**	10.49	12.97	0.920 ± 0.002	0.930
σ^2 (µm²)	Random Forest	all 8	0.939	7.65	9.57	–	0.958
σ^2 (µm²)	Random Forest (tuned)	top 4	0.936	7.98	9.86	0.934 ± 0.004	0.958
σ^2 (µm²)	**SVR (tuned)**	top 4	**0.948**	7.11	8.86	0.946 ± 0.004	0.958

Table 3: held-out test-set performance (1000 samples; MAE/RMSE in µm for d50, µm² for σ^2) and 5-fold CV on all 5000 samples (a stability check; CV mean MAE/RMSE: d50 RF 11.41/14.29, SVR 10.46/13.13; σ^2 RF 7.90/9.81, SVR 7.08/8.84).

Feature	Impurity d50	Impurity σ^2	Permutation d50	Permutation σ^2	Mean importance	Rank
Salinity	0.318	0.560	0.355	0.596	0.457	1
Temperature	0.336	0.225	0.329	0.228	0.280	2
Wave velocity	0.248	0.158	0.270	0.155	0.208	3
Refractive index	0.056	0.025	0.042	0.016	0.035	4
Tidal velocity	0.011	0.011	0.001	0.004	0.007	5
Chlorophyll-a	0.013	0.008	0.003	0.001	0.006	6
Detrital peak ratio	0.009	0.007	0.000	0.000	0.004	7
Algal peak ratio	0.009	0.006	0.000	0.000	0.004	8

Table 4: Random Forest feature importance, averaged over 10 random 70/30 splits of the training set (permutation scores normalised to sum to 1).

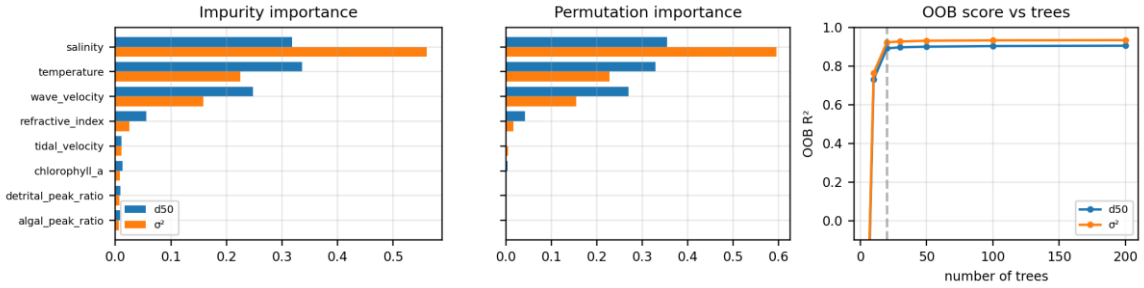


Fig. 2: "dashed line: 20 trees, where OOB R^2 levels off; the final model uses 100 trees

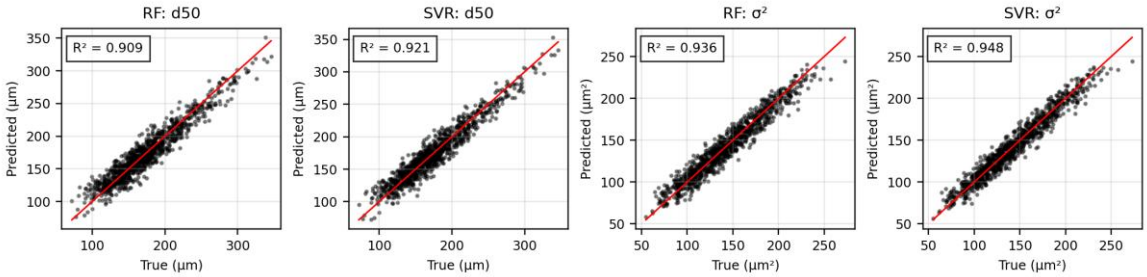


Fig. 3: test-set predictions vs ground truth (red line = perfect fit) for the tuned top-4 RF and SVR models.

6. Interpretation

- **Both models predict well; SVR is slightly better** (test R^2 0.921 vs 0.909 for d50, 0.948 vs 0.936 for σ^2, with lower MAE and RMSE). The 5-fold CV agrees (std ≤ 0.005), so this is not an artefact of one split. The targets are smooth functions, which an RBF kernel fits more naturally than piecewise-constant trees; RF flattens at the extremes in Fig. 3 because trees cannot extrapolate.
- **Feature importance.** Salinity, temperature, wave velocity and refractive index carry almost all the signal; the other four score below 0.015 (permutation ≈ 0). For σ^2 salinity dominates (impurity 0.56); for d50 salinity (0.32), temperature (0.34) and wave velocity (0.25) are comparable. Impurity and permutation rankings agree. Since this mirrors the coefficients we chose, it shows the method recovers a known structure rather than a physical finding.
- **Feature selection costs almost nothing:** dropping the four weak features lowers RF test R^2 by only 0.005 (d50) and 0.004 (σ^2). On real data this would mean expensive optical or chlorophyll sensors could be left out of a campaign if equally uninformative.
- **Near the noise ceiling.** SVR is within about 0.01 R^2 of the ceiling (0.921 vs 0.930; 0.948 vs 0.958) and RF within about 0.02: both have learned most of the structure and the remaining error is the injected noise (typical error ≈10 µm on d50, ≈6 % of its mean; ≈7 µm² on σ^2, ≈5 %).
- **Versus the paper.** Egan reports real-data test R^2 of 0.81 (d50) and 0.75 (σ^2) for RF, and 0.76 for SVR. Our higher scores come from low noise and a known relationship, so the numbers **are not comparable**; the qualitative picture (RF ≈ SVR, a few dominant features) matches.

7. Conclusions and Limitations

Random Forest and SVR both predict median particle diameter and PSD variance accurately (test R^2 0.91–0.95), top-4 feature selection loses almost nothing, and tuned SVR is marginally best. The full pipeline (held-out test set, importance-based selection, OOB and CV tuning, CV confirmation, noise-ceiling check) works as intended and is reproducible. **Limitations:** (i) the data is synthetic, so these scores show the pipeline works but **do not prove** the models predict real flocculation; (ii) inputs are independent and uniform, whereas field data is correlated and time-dependent; (iii) importance rankings reflect our chosen coefficients; (iv) the σ^2 SVR optimum lies at the edge of the C grid; (v) cross-site generalisation was not tested. **Future work:** apply the pipeline to real field data, add correlated inputs and heavier noise, extend the SVR grid, and test other sites, as the paper proposes.